

# Supplementary Materials

## BQEB: A Reproducibility-First Architecture for Modular Energy AI Benchmarking

Rakesh Kumar Agrawal

This document contains the Dataset Card, Software Card, Benchmark Card, Artifact Identifier Concordance, Reproducibility Checklist, Governance Checklist, Reviewer Checklist, Publication Checklist, Replication Protocol, and Version History accompanying the main manuscript. No content differs from the version originally reviewed; this material has been relocated, not altered, for IEEE conference length conformance.

### Appendices

#### Appendix A. Dataset Card — BQEB-Data v1

| Field                  | Value                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------|
| Name / version         | BQEB-Data v1                                                                                       |
| Size                   | 10,512 records × 37 variables                                                                      |
| Sampling interval      | 15 minutes                                                                                         |
| Coverage               | 2026-01-01 to 2026-04-20 (~109 days)                                                               |
| Nature                 | Synthetic (stochastic simulation); not field-measured                                              |
| Splits provided        | train.csv / validation.csv / test.csv (70/15/15, chronological), per IEEE DataPort package         |
| Repositories           | Harvard Dataverse (DOI 10.7910/DVN/VUVTED); IEEE DataPort (DOI 10.21227/xj9b-2885); Zenodo mirrors |
| Documented limitations | Synthetic rather than field-measured; single simulated region; simplified market assumptions       |
| Known open item        | attack_class label-generation methodology not documented in any source inspected for this paper    |

#### Appendix B. Software Card — ForecastBench Package v1.0.0

| Field                | Value                                                                                                                        |
|----------------------|------------------------------------------------------------------------------------------------------------------------------|
| Repository           | github.com/RakeshKumarAgrawal/BQEB-ForecastBench                                                                             |
| Archival DOI         | 10.5281/zenodo.21735978                                                                                                      |
| License              | MIT                                                                                                                          |
| Language / runtime   | Python 3.12                                                                                                                  |
| Test suite           | 132 automated tests, 91.25% coverage at release preparation                                                                  |
| Determinism          | Fixed random seed; deterministic preprocessing and partitioning                                                              |
| Output artifacts     | predictions.csv, metrics.csv, model_comparison.csv, dataset profiles, split files, experiment manifest — SHA-256 checksummed |
| Independent archival | Software Heritage                                                                                                            |

#### Appendix C. Benchmark Card — ForecastBench (Module 1)

| Field              | Value                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------|
| Tasks              | Next-hour electricity load forecasting; day-ahead electricity price forecasting                       |
| Models evaluated   | Linear Regression, Random Forest, Gradient Boosting (classical baselines only in the current release) |
| Metrics            | MAE, RMSE, MAPE, R <sup>2</sup>                                                                       |
| Partition          | Chronological, 70% / 15% / 15%                                                                        |
| Result values      | Not reproduced in this paper — see Section 16 and Table IV                                            |
| Peer-review status | Preprint (Research Square); not yet peer-reviewed                                                     |

#### Appendix D. Artifact Card — Identifier Concordance

This table serves as the version-concordance record recommended by this program’s internal audit process.

| Artifact                                 | Identifier                   |
|------------------------------------------|------------------------------|
| BQEB monograph                           | 10.5281/zenodo.21780521      |
| BQEB framework record (IEEE DataPort)    | 10.21227/xj9b-2885           |
| BQEB-Data v1 (Harvard Dataverse)         | 10.7910/DVN/VUVTED           |
| BQEB-Data (Zenodo mirror)                | 10.5281/zenodo.19656915      |
| BQEB-Data v1 (Zenodo mirror)             | 10.5281/zenodo.19657256      |
| BQEB-OptiData v1                         | 10.5281/zenodo.20319781      |
| ForecastBench paper (Zenodo)             | 10.5281/zenodo.19716383      |
| ForecastBench preprint (Research Square) | 10.21203/rs.3.rs-10484554/v1 |
| ForecastBench software package           | 10.5281/zenodo.21735978      |

#### Appendix E. Reproducibility Checklist

- Dataset publicly available with a persistent identifier — yes
- Data partition distributed as a concrete artifact, not only described — yes (IEEE DataPort package)
- Software publicly available under an open license — yes (MIT)
- Automated test suite present and its coverage reported — yes (132 tests, 91.25%)
- Deterministic execution (fixed seeds) — yes
- Output artifacts checksummed — yes (SHA-256)
- Correspondence between software output values and published paper tables independently confirmed — not yet completed (open item, Table IV)
- Statistical significance testing between compared models — not yet implemented (roadmap item)

#### Appendix F. Benchmark Governance Checklist

- Every artifact assigned a persistent identifier — yes
- Identifiers verified against live repository records, not copied from secondary sources — yes, for this paper
- Version-concordance table maintained — yes (Appendix D, first instance)
- Cross-artifact metric reconciliation process defined — defined; not yet executed to resolution (Table IV)
- Architecture description consistent across artifacts — partially; this paper’s account is the governing version going forward

#### Appendix G. Reviewer Checklist

- Does the manuscript claim any result it does not cite evidence for? — Reviewers should check Sections 16 and Table V against this.
- Are prior BQEB contributions clearly distinguished from this paper’s new contributions? — See Section 5 and Table I.
- Is the synthetic status of the underlying dataset disclosed prominently, not only in a limitations section? — See Section 8 and the Abstract.
- Is the unresolved benchmark-result discrepancy disclosed rather than concealed? — See Section 12 and Table IV.
- Are architecture claims (Section 7) distinguished from claims already demonstrated (Section 16)? — See Section 20.

### Appendix H. Publication Checklist

- Corresponding author and ORCID present — yes
- Funding, conflict of interest, ethics, data and software availability statements present — yes
- All artifact DOIs (dataset, software, prior papers, monograph) independently verified to resolve — yes; 9 of 17 literature references carry no DOI in the source record, and none has been invented to fill the gap
- Reference list free of fabricated citations — yes; every entry corresponds to a real, checkable source, with DOIs given where one is independently confirmed to exist
- Open governance items resolved prior to submission, if the authorial decision is to resolve rather than disclose them — outstanding; current manuscript discloses rather than resolves (editorial decision, see cover materials)

### Appendix I. Replication Protocol

An independent party wishing to verify the claims in this paper should: (1) retrieve BQEB-Data v1 from Harvard Dataverse using the DOI in Appendix D and confirm record count and variable count against Table II and Appendix A; (2) clone the ForecastBench software repository at the tagged v1.0.0 release and confirm the reported test count and coverage by running the included test suite; (3) verify the SHA-256 checksums published alongside the release artifacts against a freshly downloaded copy; (4) consult the ForecastBench paper and preprint directly for exact metric values, cross-checking the two against each other, consistent with the open item recorded in Table IV.

### Appendix J. Version History

| Artifact               | Version                     | Date                                | Identifier                   | Notes                                                                                                                                                                                 |
|------------------------|-----------------------------|-------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BQEB monograph         | v1.1.1                      | 2026                                | 10.5281/zenodo.21780521      | —                                                                                                                                                                                     |
| BQEB-Data v1           | v1                          | Released 2026-04-20 (IEEE DataPort) | 10.21227/xj9b-2885           | —                                                                                                                                                                                     |
| ForecastBench paper    | Preprint                    | Posted 2026 (Research Square)       | 10.21203/rs.3.rs-10484554/v1 | —                                                                                                                                                                                     |
| ForecastBench software | v1.0.0                      | 2026-08-01                          | 10.5281/zenodo.21735978      | —                                                                                                                                                                                     |
| This manuscript        | v1.0                        | 2026                                | Not yet assigned             | Submission Candidate; frozen for factual corrections only                                                                                                                             |
| This manuscript        | v1.1                        | 2026                                | Not yet assigned             | Minor revision: dataset table shortened, equations replaced with citation, ForecastBench relationship paragraph added                                                                 |
| This manuscript        | v1.1 (camera-ready)         | 2026                                | Not yet assigned             | Reference formatting standardized, unresolved reference placeholder removed, figures re-exported at 300 DPI, no in-text/bibliography mismatches, no tracked changes or comments found |
| This manuscript        | v1.1 (final editorial pass) | 2026                                | Not yet assigned             | Table III relabeled from historical label “Table 2b”, Acknowledgements placeholder resolved, this table reformatted for readability — no information altered                          |